Product Description

Product Information	
Product name	Single Photon SPAD Detector
Product number	SPAD-050-C-B-FC SPAD-050-C-R-FC
Manufacturer	FLIM LABS S.r.l.
Registered Address	Via della Farnesina, 3 – 00135 Rome (RM), Italy

General Description

The Single Photon SPAD Detector (SPAD) is a compact single-photon detection module designed for the detection of extremely low-intensity optical signals down to the single-photon level. It is intended for applications in fluorescence lifetime imaging microscopy (FLIM), optical spectroscopy, photon counting, and time-resolved measurements in scientific and laboratory environments.

The device converts incident optical radiation into digital electrical pulses by means of an integrated SPAD (Single Photon Avalanche Diode) sensor. Incoming photons coupled via an optical fiber are detected by the SPAD operated in Geiger mode. Each detected photon event generates a fast electrical avalanche, which is internally processed and converted into a standardized digital output pulse. The resulting photon events are available at the SMA output in LVTTL format and via the USB Type-C interface in LVDS format for direct integration with data acquisition systems.

The product is available in two variants that differ exclusively in their spectral response characteristics:

- **SPAD-050-C-B-FC**: 370 nm to 900 nm, centered at 450 nm.
- **SPAD-050-C-R-FC**: 400 nm to 1000 nm, centered at 630 nm.

Both variants integrate a thermoelectric cooling (TEC) system to stabilize detector performance and reduce dark count rate.

The device can be powered via the USB Type-C interface or by an external 9 V DC power supply using a 2.1/5.5 mm coaxial DC connector.

The module features:

- Fiber-coupled optical input via multimode FC/PC connector

- Digital SMA output (2.5 V LVTTL, 50 Ω)
- USB Type-C interface with LVDS output
- Integrated thermoelectric cooling (TEC)
- Compact aluminum enclosure suitable for system integration
- Dual power supply options (USB Type-C or external 9 V DC)

Intended Use and Misuse

Intended Use

The Single Photon SPAD Detector is intended as a compact single-photon detection module for converting incident optical radiation into digital photon event signals by professional users in research laboratories, universities, or industrial environments. It may be used as a stand-alone detector unit or integrated into custom optical and time-resolved measurement systems.

Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.